



**KAZAKH-BRITISH
TECHNICAL
UNIVERSITY**

Course name: Databases

Laboratory work 2

Submission format: word or pdf document containing screenshots of your tasks with answers that covers full-page screenshots (Alt+PrtSc).

PART 1. DDL vs DML Explanation

1. Difference between DDL and DML

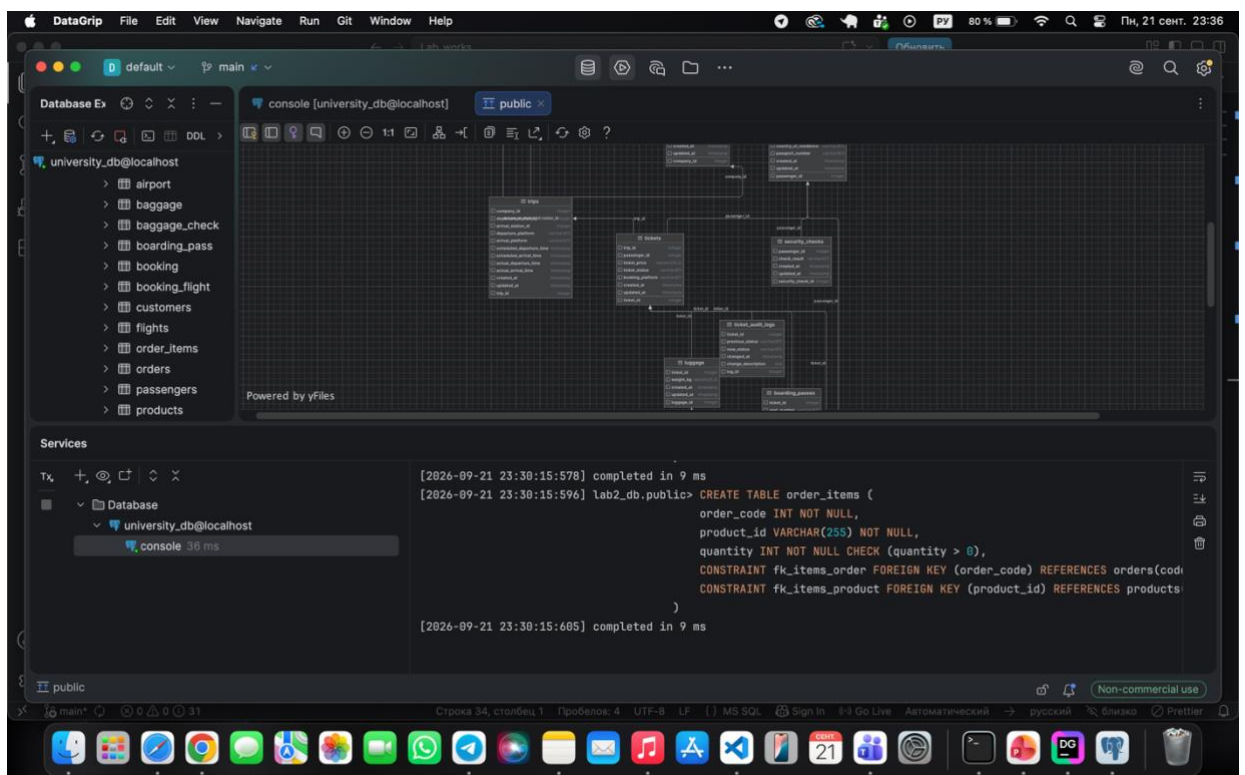
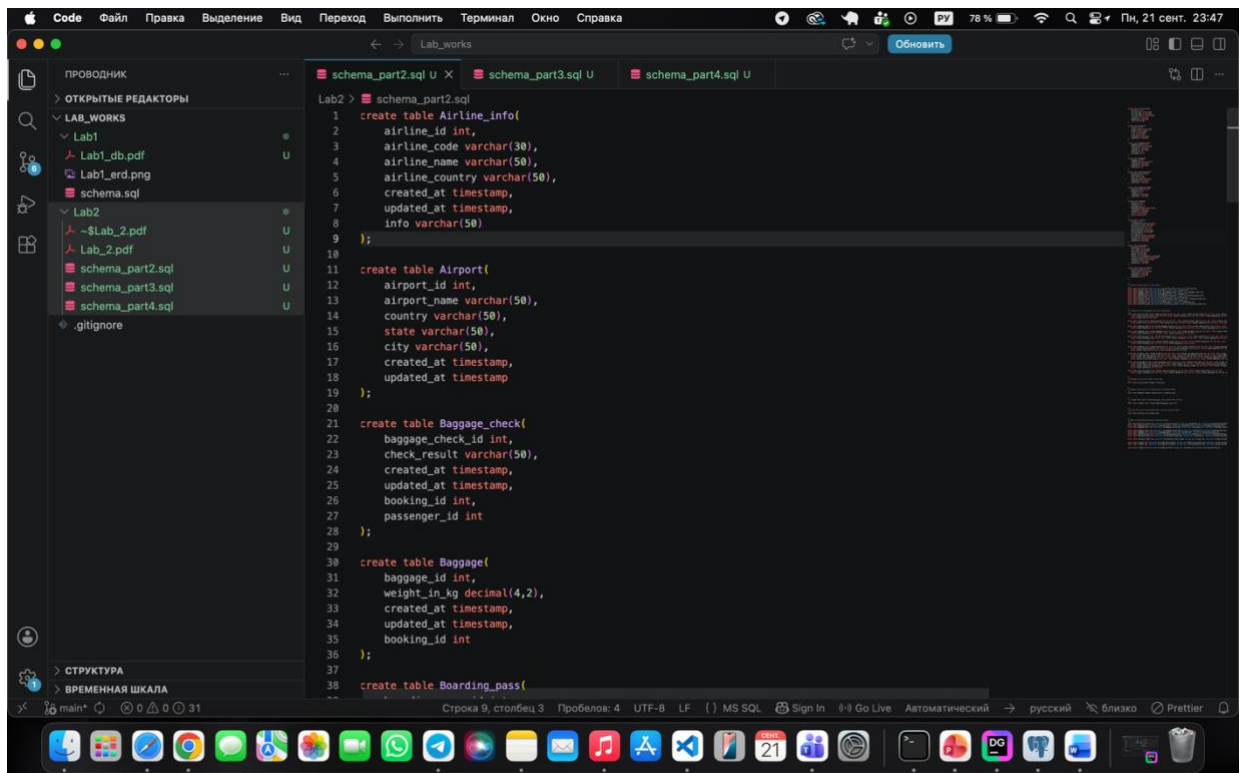
- **DDL (Data Definition Language):** используется для создания, изменения и удаления структуры базы данных (таблиц, индексов, схем и ограничений). DDL определяет то, *как* данные будут храниться.
- **DML (Data Manipulation Language):** используется для управления самими данными внутри созданных структур (добавление, чтение, изменение и удаление записей).

2. Examples of Commands

- **DDL Commands (at least 3):**
 1. CREATE — создание объектов базы данных (таблиц, схем).
 2. ALTER — изменение структуры существующей таблицы (добавление колонок, ограничений).
 3. DROP — полное удаление объектов из базы данных.
- **DML Commands (at least 4):**
 1. SELECT — выборка и чтение данных из таблиц.
 2. INSERT — добавление новых строк с данными.
 3. UPDATE — обновление существующих значений в таблицах.
 4. DELETE — удаление конкретных строк из таблицы.

PART 2. DDL

1. Create following tables with corresponding attributes:



2. Define Primary Keys for each tables;

```
Lab2 > schema_part2.sql
94 create table Security_check(
95   created_at timestamp,
96   updated_at timestamp,
97   passenger_id int
98 );
99
100
101
102
103 /*
104 2. Define Primary Keys for each tables
105 */
106 alter table Airline_info add constraint pk_airline_info primary key (airline_id);
107 alter table Airport add constraint pk_airport primary key (airport_id);
108 alter table Baggage_check add constraint pk_baggage_check primary key (baggage_check_id);
109 alter table Baggage add constraint pk_baggage primary key (baggage_id);
110 alter table Boarding_pass add constraint pk_boarding_pass primary key (boarding_pass_id);
111 alter table Booking add constraint pk_booking primary key (booking_id);
112 alter table Booking_flight add constraint pk_booking_flight primary key (booking_flight_id);
113 alter table Flights add constraint pk_flights primary key (flight_id);
114 alter table Passengers add constraint pk_passengers primary key (passenger_id);
115 alter table Security_check add constraint pk_security_check primary key (security_check_id);
116
117
118 /*
119 4. Define for all attributes not null constraint;
120 */
121 alter table Airline_info alter column airline_id set not null, alter column airline_code set not null, alter column airline_name set not null,
122   alter column airline_country set not null, alter column created_at set not null, alter column updated_at set not null,
123   alter column info set not null;
124
125 alter table Airport alter column airport_id set not null, alter column airport_name set not null, alter column country set not null,
126   alter column state set not null, alter column city set not null, alter column created_at set not null, alter column updated_at set not null;
127
128 alter table Baggage_check alter column baggage_check_id set not null, alter column check_result set not null, alter column created_at set not null,
129   alter column updated_at set not null, alter column booking_id set not null, alter column passenger_id set not null;
130
131 alter table Baggage alter column baggage_id set not null, alter column weight_in_kg set not null, alter column created_at set not null,
132   alter column updated_at set not null, alter column booking_id set not null;
133
134 alter table Boarding_pass alter column boarding_pass_id set not null, alter column booking_id set not null, alter column boarding_time set not null,
135   alter column boarding_time set not null, alter column created_at set not null, alter column updated_at set not null;
```

4. Define for all attributes not null constraint;

```
116
117
118 /*
119 4. Define for all attributes not null constraint;
120 */
121 alter table Airline_info alter column airline_id set not null, alter column airline_code set not null, alter column airline_name set not null,
122   alter column airline_country set not null, alter column created_at set not null, alter column updated_at set not null,
123   alter column info set not null;
124
125 alter table Airport alter column airport_id set not null, alter column airport_name set not null, alter column country set not null,
126   alter column state set not null, alter column city set not null, alter column created_at set not null, alter column updated_at set not null;
127
128 alter table Baggage_check alter column baggage_check_id set not null, alter column check_result set not null, alter column created_at set not null,
129   alter column updated_at set not null, alter column booking_id set not null, alter column passenger_id set not null;
130
131 alter table Baggage alter column baggage_id set not null, alter column weight_in_kg set not null, alter column created_at set not null,
132   alter column updated_at set not null, alter column booking_id set not null;
133
134 alter table Boarding_pass alter column boarding_pass_id set not null, alter column booking_id set not null, alter column boarding_time set not null,
135   alter column boarding_time set not null, alter column created_at set not null, alter column updated_at set not null;
136
137 alter table Booking alter column booking_id set not null, alter column flight_id set not null, alter column passenger_id set not null,
138   alter column booking_platform set not null, alter column created_at set not null, alter column updated_at set not null,
139   alter column status set not null, alter column price set not null;
140
141 alter table Booking_flight alter column booking_flight_id set not null, alter column booking_id set not null, alter column flight_id set not null,
142   alter column created_at set not null, alter column updated_at set not null;
143
144 alter table Flights alter column flight_id set not null, alter column sch_departure_time set not null, alter column sch_arrival_time set not null,
145   alter column departing_airport_id set not null, alter column arriving_airport_id set not null, alter column departing_gate set not null,
146   alter column arriving_gate set not null, alter column airline_id set not null, alter column act_departure_time set not null,
147   alter column act_arrival_time set not null, alter column created_at set not null, alter column updated_at set not null;
148
149 alter table Passengers alter column passenger_id set not null, alter column first_name set not null, alter column last_name set not null,
150   alter column date_of_birth set not null, alter column gender set not null, alter column country_of_citizenship set not null,
151   alter column country_of_residence set not null, alter column passport_number set not null, alter column created_at set not null,
152   alter column updated_at set not null;
153
```

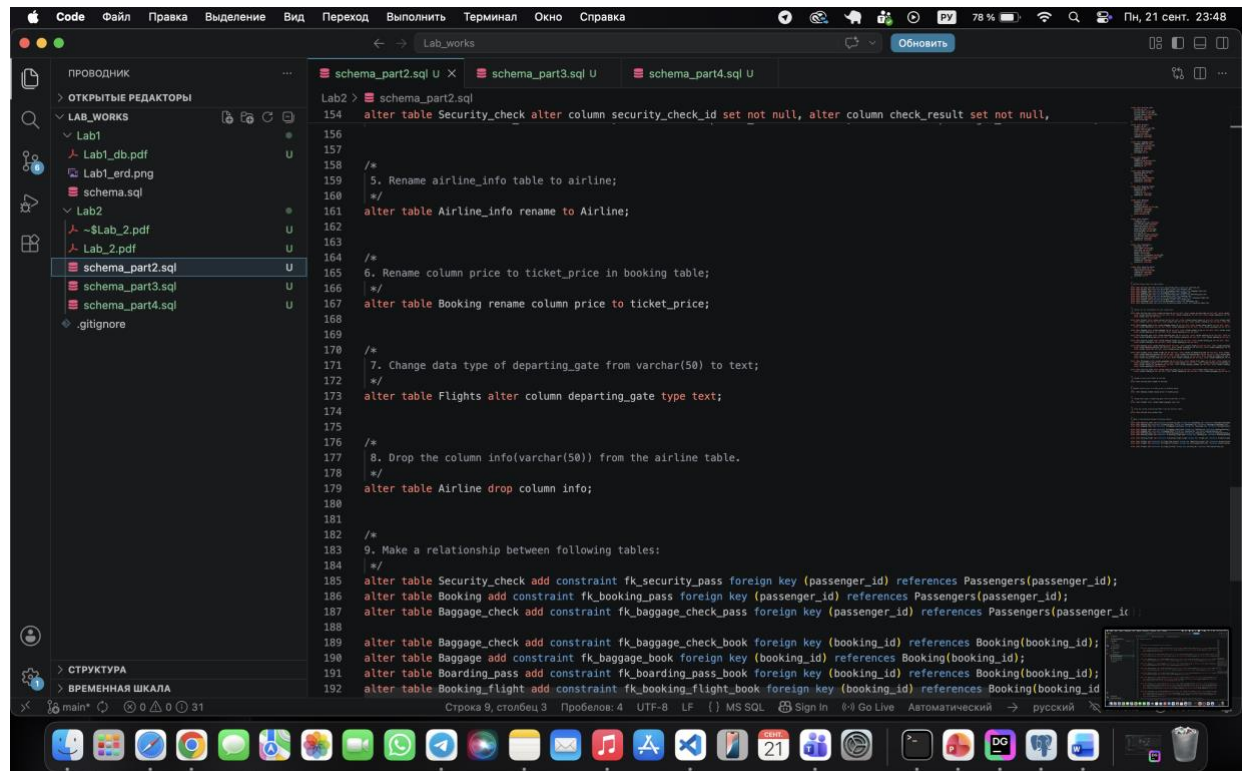
5. Rename airline_info table to airline;

6. Rename column price to ticket_price in booking table; 7. Change data type of departing_gate from varchar(50) to text;

8. Drop the column info(varchar(50)) from the airline table.

9. Make a relationship between following tables:

- Passengers with Security_check, Booking, Baggage_check by passenger_id;
- Booking with Baggage_check, Baggage, Boarding_pass, Booking_flight by booking_id;
- Flights with Booking_flight by flight_id;
- Airport with Flights by departing_airport_id;
- Airport with Flights by arriving_airport_id;
- Airline with Flights by airline_id;

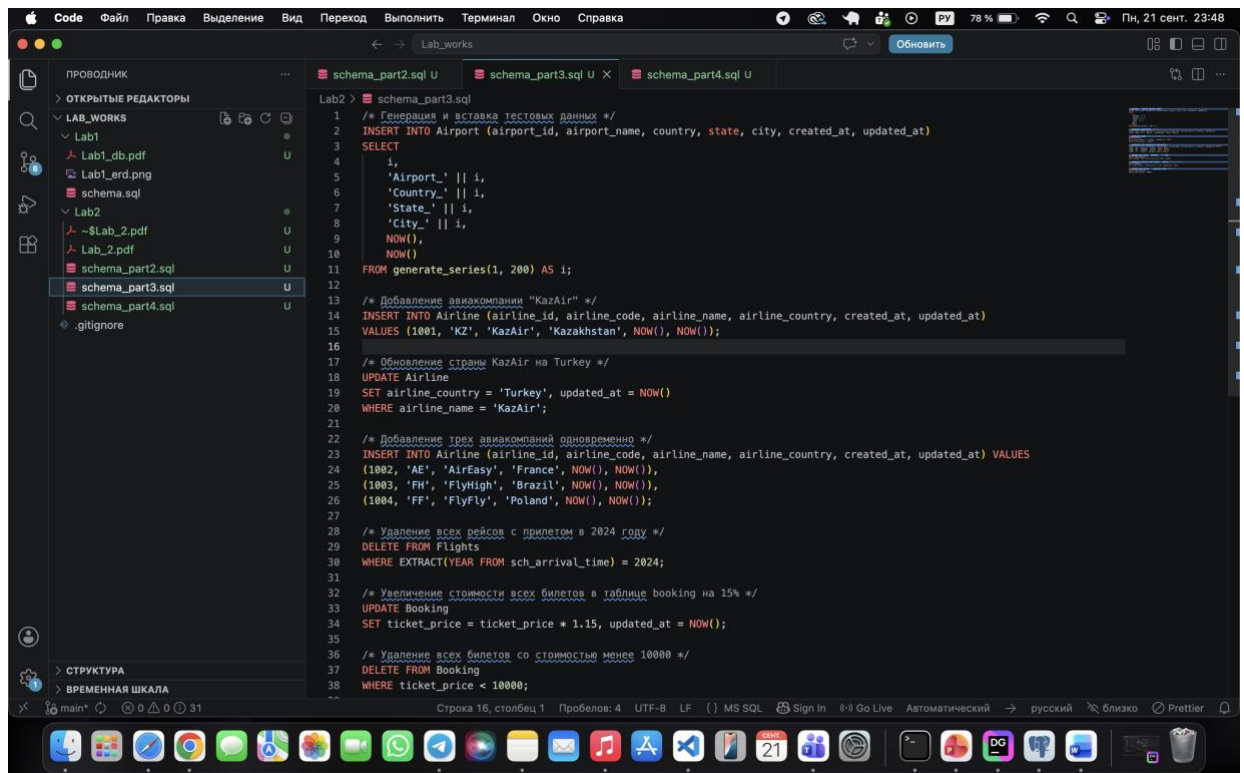


The screenshot shows a code editor with a file explorer on the left and a SQL query editor on the right. The file explorer shows a project named 'LAB_WORKS' with subfolders 'Lab1' and 'Lab2'. The 'Lab2' folder contains files 'Lab2_2.pdf', 'Lab2_2.pdf', 'schema_part2.sql', 'schema_part3.sql', 'schema_part4.sql', and '.gitignore'. The SQL query editor shows the following queries:

```
154 alter table Security_check alter column security_check_id set not null, alter column check_result set not null,
155
156 /*
157 5. Rename airline_info table to airline;
158 */
159 alter table Airline_info rename to Airline;
160
161 /*
162 6. Rename column price to ticket_price in booking table;
163 */
164 alter table Booking rename column price to ticket_price;
165
166 /*
167 7. Change data type of departing_gate from varchar(50) to text;
168 */
169 alter table Flights alter column departing_gate type text;
170
171 /*
172 8. Drop the column info(varchar(50)) from the airline table.
173 */
174 alter table Airline drop column info;
175
176 /*
177 9. Make a relationship between following tables:
178 */
179 alter table Security_check add constraint fk_security_pass foreign key (passenger_id) references Passengers(passenger_id);
180 alter table Booking add constraint fk_booking_pass foreign key (passenger_id) references Passengers(passenger_id);
181 alter table Baggage_check add constraint fk_baggage_check_pass foreign key (passenger_id) references Passengers(passenger_id);
182
183 alter table Baggage_check add constraint fk_baggage_check_book foreign key (booking_id) references Booking(booking_id);
184 alter table Baggage add constraint fk_baggage_book foreign key (booking_id) references Booking(booking_id);
185 alter table Boarding_pass add constraint fk_boarding_pass_book foreign key (booking_id) references Booking(booking_id);
186 alter table Booking_flight add constraint fk_booking_flight_book foreign key (booking_id) references Booking(booking_id);
187
```

PART 3. DML

1. Generate and insert 200 random rows in your airport database.
2. Add a new airline named "KazAir" based in "Kazakhstan" to the airline table.
3. Update the airline country "KazAir" to "Turkey".
4. Add three airlines at once: "AirEasy" in "France", "FlyHigh" in "Brazil" and "FlyFly" in "Poland".
5. Delete all flights whose arrival in 2024 year.
6. Increase the price of all tickets in booking table for flights by 15%.
7. Delete all tickets where price is less than 10000.



PART 4. Write SQL statements to create tables in the figure below

